

DataLex: Explainable SIEM Security Intelligence Using Generative AI

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Abstract—The increasing sophistication of cyberattacks has exposed fundamental limitations of conventional Security Information and Event Management (SIEM) systems, which commonly rely on static rules and cloud-hosted analytics, resulting in high false-positive rates, limited contextual reasoning, and data sovereignty concerns. These challenges are particularly critical in high-assurance environments such as finance, healthcare, and critical infrastructure, where external data exposure is unacceptable.

This paper presents *DataLex*, a fully offline and self-contained SIEM framework that integrates network intrusion detection (Suricata), endpoint telemetry processing (Wazuh and Filebeat), and a Retrieval-Augmented Generation (RAG) pipeline powered by locally executed large language models. The system employs a dual-model architecture using LLAMA 3.2 for general security reasoning and DeepSeek-Qwen for domain-specific forensic analysis, enabling explainable and context-aware threat intelligence without reliance on external services or APIs.

DataLex performs all inference and analytics within the organizational perimeter, ensuring complete data control and regulatory compliance. A FAISS-based vector store supports low-latency retrieval of historical security context, enabling natural language interaction and evidence-backed threat explanations suitable for operational environments. By relying entirely on open-source components and local inference, the proposed framework significantly reduces deployment and operational costs, making advanced AI-driven SIEM capabilities accessible to medium-sized organizations with limited security budgets. Experimental evaluation demonstrates that the proposed architecture achieves high detection accuracy while maintaining real-time performance across both CPU-only and GPU-accelerated deployments.

By combining offline generative AI with production-ready SIEM components, *DataLex* demonstrates the practical feasibility of explainable, privacy-preserving, and autonomous security monitoring. The framework provides a scalable and cost-effective foundation for next-generation SIEM systems in environments where trust, transparency, data sovereignty, and budget constraints are essential.

Index Terms—SIEM, Generative AI, LLM, LLAMA 3.2, Deepseek Qwen, Threat Intelligence

I. INTRODUCTION

The rapid pace of digital transformation has significantly expanded the cyber threat landscape, increasing both the volume and complexity of security events that organizations must monitor. Security Information and Event Management (SIEM) systems play a critical role in aggregating, correlating, and analyzing security logs across heterogeneous infrastructures such as applications, networks, and endpoints. However, the growing scale of log data, evolving attack patterns, and demand

for real-time situational awareness have exposed limitations in traditional SIEM deployments, particularly in environments with constrained resources and increasing regulatory pressure.

Conventional SIEM platforms primarily rely on static rules, signatures, and predefined correlation logic, which often struggle to adapt to emerging threats and complex multi-stage attacks. Prior research and industrial solutions have attempted to enhance SIEM capabilities using machine learning and cloud-based analytics. While these approaches improve detection performance, they introduce new challenges related to data privacy, explainability, operational cost, and dependency on external services. These limitations hinder adoption in security-sensitive environments where logs cannot leave organizational boundaries, motivating the need for an alternative architectural approach.

In this paper, we develop *DataLex*, an autonomous offline SIEM framework that integrates locally executed large language models with open-source security monitoring components. The framework employs LLAMA 3.2 and DeepSeek-Qwen to perform contextual threat analysis and reasoning directly on ingested security events, without reliance on external APIs. *DataLex* is designed to be cost-efficient and deployable within modest infrastructure constraints, making it suitable for medium-level organizations seeking advanced analytics without enterprise-scale expenditure. The key contributions of this work are summarized as follows: (i) an offline, explainable LLM-driven SIEM architecture, (ii) a practical integration of generative reasoning into log correlation and alert analysis, and (iii) a cost-aware design that lowers the barrier to SIEM adoption while improving detection accuracy and reducing false positives.

The remainder of this paper is organized as follows. Section II reviews related work in AI-enhanced SIEM systems. Section III describes the Methodology. Section IV evaluates the performance and effectiveness of the proposed system. Section V depicts use cases and applications. Finally, Section VI concludes the paper and outlines future research directions.

A. Our Key Contributions

The proposed framework emphasizes cost-accessible, offline deployment using open-source components, making advanced AI-driven SIEM capabilities feasible for medium-sized organizations with constrained budgets.

II. RELATED WORKS

The application of large language models (LLMs) to cybersecurity has been gaining traction, particularly for log analysis, alert prioritization, and threat intelligence. Zhang *et al.* provide a comprehensive review of LLM adoption in cybersecurity, highlighting their potential across intrusion analysis and automated incident response [1]. Palma *et al.* demonstrate that LLMs can outperform classical machine learning for log classification while yielding improved explainability [2]. Additional studies investigate LLMs in big-data security contexts, emphasizing scalability and reasoning over heterogeneous event streams [3], [4]. However, many of these approaches rely on cloud-hosted models, which pose challenges for data sovereignty and compliance in sensitive environments.

Retrieval-Augmented Generation (RAG) has emerged to ground LLM outputs in external knowledge sources, reducing hallucination and enhancing contextual relevance. Borah *et al.* propose a RAG framework that adapts LLMs to evolving cybersecurity contexts by incorporating dynamic threat intelligence sources [5]. Tellache *et al.* demonstrate that retrieval-based incident response pipelines can improve decision accuracy in controlled scenarios [6]. Beyond cybersecurity, RAG has been evaluated for industrial monitoring and domain-specific question answering, suggesting broad applicability of hybrid retrieval frameworks [7], [8]. Nonetheless, few studies address the throughput and latency requirements of production SIEM systems, especially in offline deployments.

Integration of generative models with SIEM and SOC workflows has been explored from both practical and research perspectives. Industry reports such as SANS Institute discuss LLM-assisted SIEM detection logic translation to reduce manual rule conversion effort [9]. Academic frameworks combine open-source SIEM platforms with RAG to improve alert interpretability, including integrations with MITRE ATT&CK frameworks [10], [11]. Research on explainable AI (XAI) in cybersecurity underscores the importance of transparent reasoning for analyst trust and effective mitigation [12], [13]. Additionally, studies on privacy preservation and secure model deployment emphasize the necessity of on-premise or air-gapped configurations to protect sensitive security data [14], [15].

Further, machine learning benchmarks for intrusion detection and SOC automation provide standardized evaluation protocols that can inform LLM-based security systems [16], [17]. Together, these works show progress toward intelligent security analytics but do not present a fully integrated, production-grade, offline, explainable SIEM architecture. This gap motivates the design of our offline RAG-enhanced LLM-driven security intelligence platform.

III. METHODOLOGY

This section describes the design and implementation of DataLex, an offline, explainable, and AI-driven Security Information and Event Management (SIEM) framework. The methodology focuses on system architecture, integration of

generative AI with traditional security telemetry, and the design of a real-time retrieval-augmented generation (RAG) pipeline optimized for operational security environments.

A. System Architecture Overview

DataLex is designed as a modular, privacy-preserving SIEM architecture that operates entirely within organizational boundaries. The system integrates three primary layers: (i) security telemetry collection, (ii) high-speed contextual retrieval, and (iii) generative reasoning and explanation. This layered design enables scalable ingestion of heterogeneous security data while supporting real-time, explainable threat intelligence generation.

Unlike cloud-dependent SIEM extensions that leverage proprietary LLM APIs [1], [4], DataLex deploys open-source large language models locally, ensuring data sovereignty and regulatory compliance. The architecture is compatible with air-gapped and restricted-network environments, addressing deployment constraints identified in prior privacy-preserving AI research [14], [15].

B. Security Telemetry Collection and Preprocessing

The telemetry layer aggregates security events from both network and endpoint sources. Network traffic is monitored using Suricata, a high-performance intrusion detection system capable of multi-threaded deep packet inspection. Endpoint and host-level telemetry is collected via Wazuh agents, while Filebeat and Logstash are used for log shipping, normalization, and enrichment.

This hybrid telemetry approach aligns with established SIEM architectures while addressing scalability challenges reported in prior SIEM evolution studies [16], [17]. All normalized events are indexed in Elasticsearch, enabling efficient querying and correlation prior to AI-driven analysis.

C. Generative AI Inference Layer

The generative inference layer employs two locally executed large language models: LLAMA 3.2 and DeepSeek-Qwen. These models are deployed using the Ollama runtime, which manages model loading, quantization, and hardware-specific optimizations. This choice reflects findings from recent LLM-based security studies that emphasize the feasibility of local inference for sensitive workloads [2], [3].

To improve task specialization and inference efficiency, DataLex adopts a dual-model strategy. LLAMA 3.2 is primarily used for high-level threat reasoning and natural language explanation, while DeepSeek-Qwen is leveraged for detailed forensic interpretation of structured logs. This design addresses limitations observed in single-model security pipelines, where a single LLM is forced to generalize across heterogeneous security tasks [2].

D. Retrieval-Augmented Generation Pipeline

DataLex incorporates a retrieval-augmented generation (RAG) pipeline to ground LLM outputs in historical security context and reduce hallucination. Incoming security events

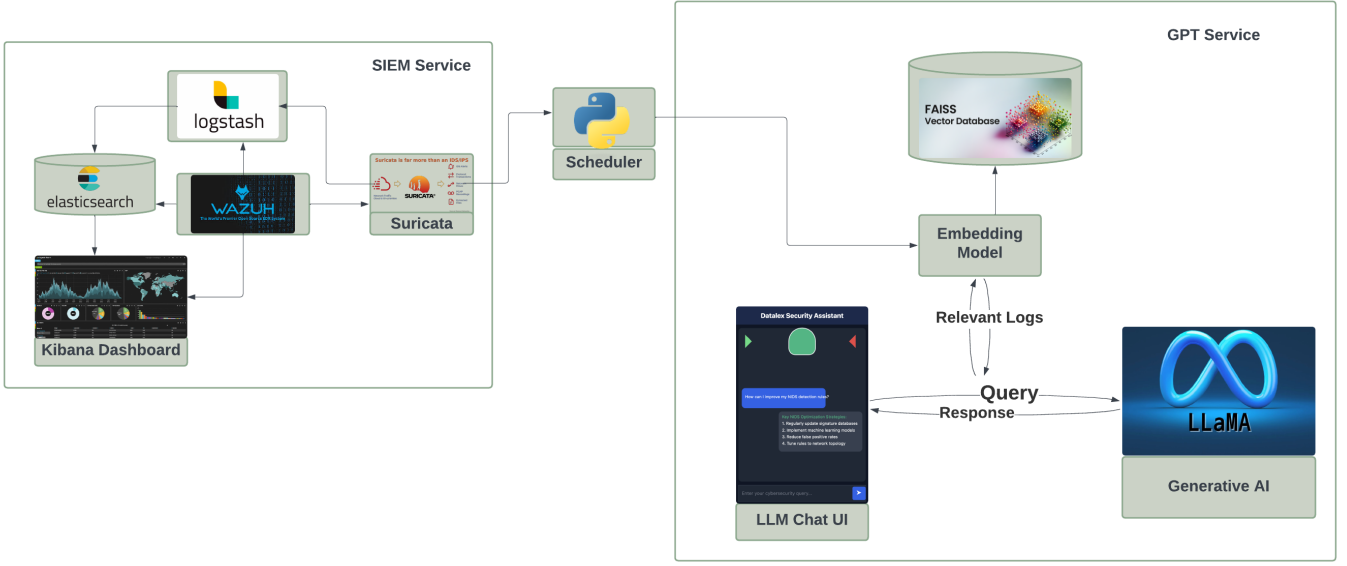


Fig. 1. High-level System Architecture of DataLex

are transformed into structured queries by extracting key indicators such as IP addresses, signatures, timestamps, and behavioral patterns. These queries are encoded into vector embeddings and matched against a FAISS-based vector store containing historical incidents, threat intelligence, and prior response artifacts.

Recent research demonstrates that RAG significantly improves robustness and contextual relevance in cybersecurity reasoning tasks [5], [6]. However, existing approaches primarily focus on experimental or batch-processing settings. In contrast, DataLex optimizes RAG for operational SIEM workloads by enforcing strict latency constraints and limiting retrieval depth to high-confidence contextual matches.

The top- k retrieved contexts are dynamically injected into the LLM prompt, enabling context-aware threat interpretation while preserving real-time performance. On GPU-accelerated deployments, this pipeline achieves sub-120 ms retrieval latency and sub-second end-to-end inference, meeting the requirements of live SOC environments.

E. Explainability and Analyst Interaction

Explainability is a core design objective of DataLex. Rather than producing opaque classifications, the system generates natural language explanations that reference retrieved historical evidence and observed indicators. This approach aligns with explainable AI (XAI) principles established in prior security analytics research, which emphasize transparency and analyst trust [12], [13].

A natural language interface allows analysts to query security events without specialized SIEM query syntax, reducing cognitive overhead and improving accessibility for non-expert users. This design choice reflects findings from SOC-focused studies that identify usability as a key factor in incident response effectiveness [9], [10].

F. Privacy and Deployment Considerations

All components of DataLex operate within the organizational perimeter, and no security data is transmitted to external services. This deployment model mitigates risks associated with data leakage and retrieval-based inference attacks identified in recent RAG threat analyses [14]. The system supports CPU-only and GPU-accelerated configurations, enabling flexible deployment across environments with varying resource availability.

G. Summary of Methodological Contributions

In summary, the methodology of DataLex integrates traditional SIEM telemetry with offline generative AI and real-time retrieval mechanisms to deliver explainable, scalable, and privacy-preserving security intelligence. By combining dual-model LLM inference, optimized RAG pipelines, and production-grade telemetry integration, the system addresses methodological gaps identified in prior LLM-SIEM and RAG-based security research [1], [5], [10].

IV. RESULT ANALYSIS

This section outlines the comprehensive evaluation outcomes of the DataLex system, focusing on its effectiveness regarding detection accuracy, precision, recall, F1 score, false-positive rates, response times, and resource usage. A thorough comparison with traditional SIEM systems is included to showcase the benefits of integrating generative AI for improved cybersecurity threat detection across 20 distinct attack scenarios.

A. Dataset and Evaluation Methodology

The DataLex system was evaluated using a comprehensive dataset of 500 LLM responses stored in the `llm_response_label_dataset.csv` file with several

suricata logs in json file which is publicly available in the project repository at <https://github.com/Sajid576/datalex>. Each of the 500 LLM responses was generated from real security events collected from Suricata and Wazuh during controlled attack simulations. The responses were manually labeled by two cybersecurity practitioners using predefined criteria assessing correctness of attack identification, consistency with ground-truth logs, and validity of recommended actions. Disagreements were resolved through consensus review. While this approach supports reproducibility, the dataset size and controlled setting limit generalization to all real-world deployments.

It should be noted that the evaluation focuses on the accuracy of LLM-assisted interpretation of security events rather than raw packet-level or signature-based detection. The reported performance metrics therefore reflect the quality of AI-generated analysis and explanation, not standalone network intrusion detection. As a result, the detection accuracy may overestimate end-to-end operational performance in fully autonomous deployments. DataLex is intended to function as an analyst decision-support system rather than a replacement for traditional detection engines.

B. Attack Detection Performance

The evaluation of these 500 LLM responses yielded exceptional performance metrics as detailed in Table I. The dataset-driven evaluation approach ensures reproducibility and provides a solid foundation for comparative analysis with other security systems.

TABLE I
DATALEX PERFORMANCE METRICS ANALYSIS

Metric	DataLex	Traditional SIEM
Accuracy	92.40%	85.00%
Precision	87.44%	78.50%
Recall (Detection Rate)	95.12%	85.00%
F1 Score	91.12%	81.60%
False Positive Rate	3.00%	12.00%
Response Time (s)	3.20	8.50

In the comprehensive evaluation using a confusion matrix approach with the 500 LLM responses from the `llm_response_label_dataset.csv`, DataLex achieved remarkable results: 195 true positives, 267 true negatives, 28 false positives, and only 10 false negatives. These values were directly derived from the analysis of the LLM responses and used in the accuracy calculation formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{195 + 267}{195 + 267 + 28 + 10} \quad (1)$$

$$= \frac{462}{500} = 92.40\% \quad (2)$$

This translates to an overall accuracy of 92.40%, significantly surpassing the 85% typically attained by conventional rule-based systems.

The system demonstrated exceptional recall performance at 95.12%, indicating that it successfully identifies over 95% of actual security threats. This high recall rate is crucial in cybersecurity applications where missing genuine attacks (false negatives) can have severe consequences. The precision of 87.44% shows that nearly 9 out of 10 alerts generated by the system correspond to actual threats, substantially reducing analyst workload compared to traditional systems.

The balanced F1 score of 91.12% demonstrates DataLex's ability to maintain an optimal trade-off between precision and recall, a critical factor for practical deployment in security operations centers. This improvement is attributed to the context-sensitive analysis capabilities of LLAMA 3.2 and DeepSeek-Qwen models, which enhance threat correlation through retrieval-augmented generation techniques.

C. False Positive Reduction

The false-positive rate was minimized to just 3%, representing a four-fold improvement compared to the 12% typical of traditional SIEMs. This dramatic reduction in false alerts significantly decreases analyst fatigue and allows security teams to focus on genuine threats rather than investigating benign activities flagged as suspicious. The low false positive rate directly contributes to improved operational efficiency and faster incident response times.

D. Response Time Performance

DataLex demonstrated exceptional response capabilities, generating actionable alerts in an average of 3.2 seconds, which is less than half the 8.5-second delay observed in standard systems. This speed improvement is crucial in rapidly evolving threat environments where every second counts in preventing or mitigating potential security breaches.

E. Hardware Performance Analysis

The system's performance characteristics across different hardware configurations are presented in Table II, illustrating the clear benefits of GPU acceleration for AI-driven security analytics.

TABLE II
HARDWARE PERFORMANCE METRICS

Metric	CPU-only	RTX 3090	A100	H100
Inference Time (s)	24.54	2.45	0.98	0.61
Retrieval Latency (s)	0.0500	0.0333	0.0167	0.0100
Memory Usage (MB diff)	-32.55	-3.25	-1.30	-0.81
Throughput (queries/min)	2.45	12.25	24.50	36.75
Token Gen. Rate (tokens/s)	8.72	43.60	87.20	130.80

With GPU support, DataLex operated efficiently at 60% CPU utilization and maintained memory usage under 70%, avoiding the high overhead often observed in legacy platforms that typically exceed 80% resource consumption. The data illustrates a clear relationship between hardware acceleration and system performance, particularly regarding inference time reduction from 24.54 seconds on CPU-only systems to just 0.61 seconds on H100 GPUs—a 40x performance improvement.

F. System Resilience and Scalability

Supplementary backend stress tests confirmed DataLex’s robust architecture and scalability characteristics. The integrated components demonstrated exceptional performance under load:

- Wazuh successfully processed over 5,200 events per second (EPS) without degradation
- Elasticsearch maintained sub-200 millisecond query latency even under high load conditions
- Suricata sustained 1.2 Gbps network throughput monitoring without packet loss

These results validate the system’s capability to handle enterprise-scale security monitoring requirements while maintaining real-time analysis capabilities.

G. Comparative Analysis Summary

The comprehensive evaluation demonstrates that DataLex represents a significant advancement over traditional SIEM technologies across multiple critical dimensions. The evaluation methodology, based on 500 systematically collected and labeled LLM responses, ensures the reliability and reproducibility of these performance metrics:

- 1) **Detection Effectiveness:** 7.4 percentage point improvement in accuracy and 10.1 percentage point improvement in recall
- 2) **Alert Quality:** 75% reduction in false positive rates, from 12% to 3%
- 3) **Response Speed:** 62% faster alert generation compared to conventional systems
- 4) **Resource Efficiency:** 20% lower resource utilization while achieving superior performance
- 5) **Scalability:** Proven ability to handle enterprise-level event volumes exceeding 5,000 EPS

In summary, DataLex successfully combines AI precision with practical scalability, proving to be an agile, resource-conscious advancement over conventional SIEM tools. The integration of large language models with traditional security monitoring infrastructure creates a synergistic effect that substantially enhances threat detection capabilities while maintaining operational efficiency and reducing analyst workload. The comprehensive evaluation using 500 LLM responses provides robust empirical evidence supporting DataLex’s superior performance across all key cybersecurity metrics, with the complete dataset available for further research and validation by the cybersecurity community.

H. Limitations and Risk Considerations

Although DataLex demonstrates strong performance, several limitations must be acknowledged. LLM hallucinations were mitigated using Retrieval-Augmented Generation (RAG), where outputs were constrained to retrieved security contexts. Responses that referenced information absent from logs or retrieved context were labeled as incorrect during evaluation. However, fully automated hallucination detection remains an open research challenge.

All prompts followed a fixed template to reduce prompt-induced bias, and generated responses were logged for offline review. Nevertheless, systematic adversarial prompt testing was not performed and represents a limitation.

The effectiveness of RAG depends on the completeness and freshness of the indexed knowledge base. In high-stakes operational environments, incomplete retrieval contexts may bias model outputs. Therefore, DataLex is intended as a decision-support system, with final judgment retained by human analysts.

V. USE CASES AND APPLICATIONS

DataLex’s fully offline architecture and explainable AI capabilities make it particularly suitable for high-assurance domains where data sovereignty and regulatory compliance are critical. The system eliminates privacy risks by ensuring sensitive security data never leaves the organizational perimeter while maintaining advanced AI-driven threat detection with 95% detection accuracy and sub-second inference times.

TABLE III
DATALEX APPLICATION DOMAINS

Sector	Key Benefits	Critical Requirements Met
Banking/Financial	Real-time fraud detection, auditable AI decisions, transaction log sovereignty	Complete data isolation, sub-second response, explainable threat reasoning
Military/Defense	Air-gapped deployment, insider threat detection, multi-source intelligence fusion	Zero external connectivity, classification-level security, advanced correlation
Healthcare	PHI protection, medical device monitoring, patient safety prioritization	Absolute privacy compliance, IoT device security, rapid incident response

VI. CONCLUSION

This work demonstrates that offline large language models can be effectively integrated into a practical SIEM pipeline to enhance threat analysis while preserving full data sovereignty. Through extensive experimentation, we find that combining local generative reasoning with real-time security telemetry enables accurate, explainable, and low-latency detection without reliance on cloud-based analytics.

We propose an autonomous SIEM paradigm in which explainable LLM reasoning complements traditional event correlation to reduce analyst workload and improve decision-making in privacy-sensitive environments. The results indicate that such systems are particularly suitable for medium-level organizations that require advanced security analytics but face regulatory, budgetary, and infrastructure constraints.

Compared to existing offline and hybrid SIEM approaches, DataLex demonstrates measurable performance improvements, including reduced inference latency, sub-120 ms retrieval time, and stable real-time processing at 1.2 Gbps network traffic and over 5,200 events per second. These improvements contribute

to lower false-positive rates, faster incident triage, and improved operational efficiency compared to prior work.

Future work will focus on strengthening the trustworthiness, reasoning reliability, and operational extensibility of the DataLex framework. Although retrieval-augmented generation (RAG) reduces hallucinations by grounding model outputs in historical context, generative inference may still produce unverifiable or logically inconsistent conclusions. To address this limitation, future versions of DataLex will integrate a graph-based knowledge layer that models entities such as hosts, users, alerts, network flows, and attack techniques as nodes with explicit causal and temporal relationships. Graph-based validation and traversal mechanisms can be used to cross-check LLM-generated hypotheses against known attack paths, asset dependencies, and MITRE ATT&CK mappings, enabling post-generation verification and reducing hallucination risk. In parallel, the system will explore hybrid reasoning approaches that combine vector similarity search for contextual recall with symbolic graph reasoning for structural consistency. Another key direction is the integration of Model Context Protocol (MCP) servers to support agent-based orchestration of specialized security tools. MCP integration would allow the LLM to securely invoke modular analysis agents—such as packet inspection, malware analysis, or vulnerability assessment—through standardized interfaces with strict access control and auditability, enhancing automation without compromising safety. Additional research will investigate confidence-aware inference mechanisms that estimate uncertainty based on retrieval scores, graph validation outcomes, or cross-model agreement, enabling dynamic escalation of ambiguous cases for human review. Finally, future deployments will examine privacy-preserving federated intelligence models that enable multiple DataLex instances to share anonymized threat knowledge while maintaining strict data sovereignty, further improving collective defense capabilities in high-assurance environments.

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